LABSHEET 07

Primary Syllabus Exercise Reference:

Develop a complete Django app mapping model array collections out to unordered item matrices (Fruits)

alongside ordered indices (Selected Event Students).

Task 7.1: Implement robust fallback condition logic filters inside the Django template layer ({% if %}) to show an alert if arrays arrive empty.

```
from django.http import HttpResponse
def home(request):
    fruits = ["Apple", "Banana", "Mango"]
    students = ["Shreya", "Rahul", "Priya"]

    return HttpResponse(f"""
    <html>
    <body>
        <h2>Fruits</h2>
        {"<ul>" + "".join(f"<li>{x}</li>" for x in fruits) + "</ul>"
        if fruits else "<p>No fruits available.</p>"}

        <h2>Selected Event Students</h2>
        {"<ol>" + "".join(f"<li>{x}</li>" for x in students) + "</ol>"
        if students else "<p>No students selected.</p>"}
    </body>
    </html>
    """)

# urls.py:
# path("", home, name="home")
```

Task 7.2: Incorporate dynamic table sorting tools inside the template views to order records based on specific field properties (e.g., student name).

```
from django.http import HttpResponse
def home(request):
    students = [
        {"name": "Shreya", "roll": 101},
        {"name": "Rahul", "roll": 103},
        {"name": "Ananya", "roll": 102}
    ]

    # Sort by student name
    students = sorted(students, key=lambda x: x["name"])

    html = "<h2>Selected Students</h2>"
    html += "<table border='1'><tr><th>Name</th><th>Roll No</th></tr>"

    for s in students:
        html += f"<tr><td>{s['name']}</td><td>{s['roll']}</td></tr>"

    html += "</table>"
    return HttpResponse(html)
```

Task 8.3: Build a simple user search input form that screens long array matrices and subsets lists dynamically using back-end request processing rules.

```
from django.http import HttpResponse
def home(request):
    students = ["Shreya", "Rahul", "Ananya", "Priya", "Arjun"]

    search = request.GET.get("search", "")
    result = [s for s in students if search.lower() in s.lower()]
```

```
html = """
<h2>Student Search</h2>
<form>
    <input type="text" name="search" placeholder="Search student">
    <button type="submit">Search</button>
</form>
<ul>
"""

for student in result:
    html += f"<li>{student}</li>"

html += "</ul>"

if not result:
    html += "<p>No student found.</p>"

return HttpResponse(html)
```